(ii) $\frac{2^8 \times 3^5 \times 5^4}{3^3 \times 5^3 \times 2^4} = \frac{2^8}{2^4} \times \frac{3^5}{3^3} \times \frac{5^4}{5^3}$ [grouping exponential numbers with the same base]

$= 2^{8-4} \times 3^{5-3} \times 5^{4-3} = 2^4 \times 3^2 \times 5^1 \quad \left[\frac{a^m}{a^n} = a^{m-n}\right]$

(iii) $\frac{6^4}{6^0} = 6^{4-0} = 6^4$ (or) $\frac{6^4}{6^0} = \frac{6^4}{1} = 6^4$ [since $6^0 = 1$]

Try these

Simply the following.

1. $23^5 \div 23^2$
2. $11^6 \div 11^3$
3. $(-5)^3 \div (-5)^2$
4. $7^3 \div 7^3$
5. $15^4 \div 15$

3.3.3 Power of Exponential form

Let us find the value of $\left(2^2\right)^5$

$\left(2^2\right)^5 = 2^2 \times 2^2 \times 2^2 \times 2^2 \times 2^2 = 2^{2+2+2+2+2}$ (By Product rule)

$2^{10} = 2^{2\times5}$

Similarly, $\left(3^3\right)^4 = 3^3 \times 3^3 \times 3^3 \times 3^3$

$= 3^{3+3+3+3} = 3^{12} = 3^{3\times4}$

$\left(5^6\right)^2 = 5^6 \times 5^6 = 5^{6+6} = 5^{12} = 5^{6\times2}$

Try these

Simplify and write the following in exponent form.

1. $\left(3^2\right)^3$
2. $\left[(-5)^3\right]^2$
3. $\left(20^6\right)^2$
4. $\left(10^3\right)^5$

In general, for any non-zero integer 'a' and whole number 'm' and 'n',

$\left(a^m\right)^n = (a^m \times a^m \times a^m \ldots \times a^m)$ (n times)

$= a^{m+m+m\ldots+m}$ (n times)

$= a^{m\times n}$

Hence, $\left(a^m\right)^n = a^{m\times n}$

This is called Power Rule of exponents.



Fig. 3.5

Example 3.9 Simplify using power rule of exponents.

(i) $\left(8^3\right)^4$ (ii) $\left(11^5\right)^2$ (iii) $\left(2^6\right)^2 \times \left(2^4\right)^7$

Solution

(i) $\left(8^3\right)^4 = 8^{3\times4} = 8^{12}$ [since $\left(a^m\right)^n = a^{m\times n}$]

(ii) $\left(11^5\right)^2 = 11^{5\times2} = 11^{10}$ [since $\left(a^m\right)^n = a^{m\times n}$]

(iii) $\left(2^6\right)^2 \times \left(2^4\right)^7 = 2^{6\times2} \times 2^{4\times7}$ [since $\left(a^m\right)^n = a^{m\times n}$]

$= 2^{12} \times 2^{28}$

$= 2^{12+28} = 2^{40}$ [since $a^m \times a^n = a^{m+n}$]

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